

Incident Management & Human Approval Workflow

Lesson 5 Assessment — n8n Production-Style Automation Documentation

Purpose: Document the design, routing, alert throttling, notification, SLA escalation, human approval, decision processing, testing, and submission structure required for the assessment.

Assessment scope: Incident triggering → severity routing → duplicate suppression → Slack notification → SLA wait → Twilio SMS escalation → interactive Slack approval → webhook resume → Approved/Rejected paths.

1. Workflow Architecture

Stage	Purpose	Expected n8n components
1. Incident Trigger	Accept an incident event and establish the incident payload.	Manual Trigger or Webhook Trigger
2. Throttle Check	Suppress repeated alerts within the configured time window.	Code node / static data or equivalent
3. Priority Routing	Separate CRITICAL incidents from INFO events.	Switch or IF
4. Slack Alert	Notify stakeholders with Incident ID, message, and timestamp.	Slack
5. SLA Wait	Pause for the required acknowledgement window.	Wait — 10 min; 30 sec for testing
6. Escalation	Escalate unresolved incidents to the on-call recipient.	Twilio SMS
7. Human Approval	Request an explicit Approve/Reject decision.	Slack interactive approval
8. Resume Callback	Receive the decision through the required POST endpoint.	Webhook — slack-approval-response
9. Decision Routing	Continue through Approved or Rejected execution paths.	IF / Switch

2. Incident Notification Flow

An incident enters through a Manual Trigger or Webhook Trigger. The workflow first evaluates whether the event is eligible for notification. If the event is a duplicate within the configured throttle window, the alert is suppressed. Eligible incidents are then routed by severity. CRITICAL incidents follow the operational notification and escalation path, while INFO events follow the lower-priority notification path.

Recommended incident payload

incidentId: INC-001

severity: CRITICAL or INFO

message: Human-readable incident description

acknowledged: true/false

onCallPhone: designated escalation recipient

timestamp: event timestamp

3. Alert Throttling & Duplicate Suppression

The workflow maintains alert state using workflow static data or an equivalent n8n mechanism. The Incident ID is used as the primary duplicate key. When a matching incident has already generated an alert inside the configured time window, the new alert is suppressed. Otherwise, the workflow records the alert timestamp and continues to notification.

Logic: Receive incident → read stored alert state → compare current time with previous alert time → if inside throttle window, suppress → otherwise update state and continue to Slack.

Validation evidence: Run the same incident twice within the throttle window. The first execution should generate the eligible alert; the repeated event should follow the duplicate-suppression path.

4. Slack Notification Configuration

The immediate Slack notification should clearly identify the incident and provide enough context for an operator to act. The assessment specifically requires the Incident ID, message, and timestamp.

Field	Example
Incident ID	INC-001
Severity	CRITICAL
Message	Payment API returning 500 errors
Timestamp	2026-08-26T12:00:00Z
Status	Unacknowledged

Credential note: A real Slack credential is required for live Slack delivery and interactive approval testing. If the credential is unavailable during development, the Slack branch can be prepared and tested with an approved test/mock configuration, but the final evidence should clearly state the limitation.

5. SLA Waiting & SMS Escalation Design

After the initial notification, the workflow pauses using an n8n Wait node. The assessment requires a 10-minute SLA wait period. For development/testing, 30 seconds may be used as a test substitution. When the wait completes, the workflow checks whether the incident remains unacknowledged. If escalation is required, Twilio sends an SMS to the designated on-call recipient.

Condition	Action
Incident acknowledged before SLA expires	Do not escalate; continue according to the designed acknowledgement path.
Incident remains unacknowledged after SLA	Send Twilio SMS escalation to the designated on-call recipient.
Testing mode	Use 30-second Wait instead of 10 minutes and document the substitution.

6. Human Approval Workflow

After the operational escalation/notification stage, the workflow requests a human decision. The Slack approval interaction should provide two explicit outcomes: Approve and Reject. The decision is returned through a POST webhook callback using the required endpoint name **slack-approval-response**. The workflow then resumes and routes the response into the corresponding Approved or Rejected path.

Approval sequence

Incident → Approval request → Human selects Approve/Reject → POST callback → Webhook receives response → Decision processing → Approved path OR Rejected path.

Decision	Workflow behavior
Approved	Route to Approved execution path and record/process the approval result.
Rejected	Route to Rejected execution path and notify/process the rejection result.
Invalid/missing decision	Use validation/fallback handling rather than silently treating it as approved.

7. Webhook Resume Endpoint

Configure a POST Webhook node using the endpoint name **slack-approval-response**. The callback payload should contain enough information to correlate the response with the incident and identify the decision. The workflow should validate the incoming decision before routing.

Example callback data: incidentId = INC-001; decision = APPROVED; responder = on-call-user; timestamp = callback time.

8. Node Naming & Documentation Standards

Node	Production-style name
Trigger	Trigger_Incident_Manual
Webhook intake	Webhook_Incident_Intake
Throttle	Check_Alert_Throttle
Severity router	Route_By_Severity
Slack alert	Notify_Slack_Incident
SLA wait	Wait_SLA_10min
Acknowledgement check	Check_Acknowledged
Twilio	Escalate_Twilio_SMS
Approval request	Request_Slack_Approval
Callback	Webhook_Slack_Approval_Response
Decision	Route_Approval_Decision
Approved path	Process_Approved
Rejected path	Notify_Rejected

Each major stage should include an n8n annotation describing its purpose, expected input/output, operational assumptions, and any test substitution or limitation.

9. End-to-End Test Plan

Test	Expected result	Evidence
CRITICAL incident	Critical routing and immediate Slack alert path execute.	Execution screenshot
INFO incident	INFO routing path executes without critical escalation behavior.	Execution screenshot
Duplicate incident	Repeated alert inside throttle window is suppressed.	Two execution results
SLA acknowledgement	Acknowledged incident avoids unnecessary escalation.	Execution evidence
SLA timeout	Unacknowledged incident reaches Twilio escalation.	Twilio/Execution evidence
Approval	Approve callback resumes workflow and reaches Approved path.	Slack + execution evidence
Rejection	Reject callback resumes workflow and reaches Rejected path.	Slack + execution evidence

10. Assessment Deliverables

- Loom or YouTube demonstration video showing incident triggering, Slack alerting, throttling, SLA Wait behavior, SMS escalation, interactive approval, webhook resume, and Approved/Rejected processing.
- Exported n8n workflow JSON containing the required trigger, throttle, routing, Slack, Wait, Twilio, approval, webhook, decision, and annotation components.
- PDF/DOCX documentation package covering workflow architecture, notification flow, throttling, Slack configuration, SLA escalation, approval workflow, and naming/documentation standards.
- Workflow screenshots / execution evidence for the main notification, escalation, throttling, and approval paths.
- Public GitHub repository or another accessible source location for the workflow and supporting materials.
- Description/additional comments explaining challenges, workarounds, assumptions, limitations, test substitutions, or assessor notes.

11. Submission Notes / Assumptions

This document is structured directly from the Lesson 5 assessment requirements. Live Slack and Twilio delivery depend on available credentials/test configurations. If a real integration cannot be enabled, the implementation should document the test substitution or limitation rather than presenting simulated evidence as live delivery.

Candidate: _____

Workflow: Incident Management & Human Approval

Loom/YouTube: _____

GitHub: _____

Submission date: _____